QPCode **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Exam Type EXAMINATION Month YYYY

Semester SEMESTER

DSE-MG4DSEPHY201**- NUMERICAL METHODS FOR COMPUTATIONAL PHYSICS**

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. Write an example of an algebraic equation and a transcendental equation. [U] / [1]

2. What are the steps involved in the Gauss Elimination method? [K] / [1]

3. Mention two limitations of the Newton-Raphson method. [U] / [1]

4. Why is the Gauss-Seidel method known as the method of successive displacements? [K] / [1]

5. Write the normal equations to fit a straight line of the form y=mx+c. [U] / [1]

6. Create a difference table for [U] / [1]

X	0.1	0.2	0.3	0.4	0.5
Y	1.40	1.56	1.76	2.00	2.28

7. Write the difference between Newton's backward difference formula and the divided difference formula. [U] / [1]

8. If $D = \frac{1}{h}\left(\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} - - -\right)$ where D is the differential operator and Δ is the forward difference operator, how can you find the second derivative from a table of given values? [A] / [2]

9. Prove that $\mu + \frac{\delta}{2} = E^{1/2}$ [A] / [2]

10. Write Simpson's 3/8 rule. [K] / [1]

11. When can we say that the solution of the finite difference equation is stable? [U] / [1]

12. What is the process of discretisation in the finite difference method for solving the diffusion equation? [U] / [2]

13. Give two features of RK method. [U] / [1]

14. Given $\frac{dy}{dt} = \frac{y-t}{y+t}$ with initial condition y=1 at t=0. Find y(0.1) by taking a step size of 0.05 and using the Euler formula $y_1=y_0 + h\, f(t_0,y_0)$. [U] / [2]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. Solve $x^3-9x+1=0$ for the root between 2.5 and 3 by the bisection method. Do at least 3 iterations. [U] / [1]

16. Find the inverse of $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 4 \\ 2 & 4 & 7 \end{bmatrix}$ by Gauss-Jordan method. [U] / [1]

17. Write the algorithm to find the largest eigenvalue by the power method. [A] / [2]

18. Using Newton's forward difference formula , find a cubic polynomial in x which takes on the values -3, 3, 11, 27 and 57, when x=0, 1, 2, 3 and 4, respectively. [U] / [1]

19. Find the approximate value of $y = \int_0^{\pi} \sin x \, dx$ using trapezoidal rule by dividing the entire range into 4 equal parts. [A] / [2]

20. A missile is launched from a ground station. The acceleration during its first 80 s of the light, as recorded, is given in the following table. Compute the velocity of the missile when t=80 s, using Simpson's rule. [E] / [2]

Time (s)	0	20	40	60	80
Acceleration (m/s^2)	30	33.34	37.75	43.25	50.67

21. Solve the following initial boundary value problem using Schmidt method. $\frac{\partial T}{\partial t} = \frac{\partial^2 T}{\partial x^2}, 0 \leq x \leq 1$; given $T = \sin \pi x \; when \; t = 0, 0 \leq x \leq 1 \; and \; T = 0 \; at \; x = 0 \; and \; x = 1 \; for \; t > 0$. Choose Δx=0.2 and hence find the numerical solution at t=0.002 [E] / [2]

22. Give the graphical explanation to the modified Euler's method. [An] / [2]

[5x6 = 30]